

HYBRID EMULATION

And The Challenges Ahead



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Hybrid emulation is a technique that combines emulation and virtual prototyping to enable earlier architecture optimisation and software development, as well as higher performance for software-driven register-transfer level (RTL) verification even when critical IP RTL is not available. Hybrid emulation is done when part of the system is run in the emulator and the other part is run in a virtual prototype.

Hybrid emulation with a high-performance emulator can run at multi-megahertz speeds, fast enough to provide a productive debugging experience. When using the hybrid emulation system for software development, you require access to full-visibility debug features of the emulator, where some of the design is modelled. This allows a full system-level view for verification.

Hybrid emulation enables system on chip (SoC) development teams to take full advantage of their investments in a high-performance emulation system across their entire project to better meet increasing software, verification and competitive challenges. Typically, a model of the processor is run in the virtual platform and then the rest of the design is modelled by running the RTL on the emulator.

Benefits of hybrid emulation

Hybrid emulation offers several benefits for SoC architecture, software and verification teams. For architects, it provides a mechanism to implement key elements of the SoC that are only available as RTL in a high-performance, cycle-accurate manner to support their architecture optimisation efforts. Users can choose cycle-accurate, approximately-timed or loosely-timed synchronisation between the virtual prototype and the emulator to make speed/accuracy tradeoffs.

Three main uses for hybrid emulation are architectural validation with accurate models, early software development and hardware verification.

In architectural validation with accurate models, high-speed models used to model processors are only functionally-accurate and not cycle-accurate, and you cannot get accurate performance or accurate performance data from these at that level of detail. Of course, you could just run the RTL of the processor on a simulator, but that is too slow.

In early software development, transactional-level models are often not available, are too low or do not exactly match the RTL for the block. But a hybrid approach with the processor in a virtual platform and all or most of the rest of the design in the emulator is fast enough for productive software development.

Hardware verification allows running actual software load.

Architectural validation. For SoC architecture validation, you, as a user, have to link a high-performance emulator like ZeBu Server-3 to a virtual prototype exploration tool like Platform Architect MCO. ZeBu Server is the industry's fastest emulation system. ZeBu solution includes ZeBu Server-3 emulator and a broad portfolio of ZeBu transactors, memory models and ZEMI-3 transactor compiler for rapid development of virtual system-level verification environments.

Fig. 1: Virtual prototyping

